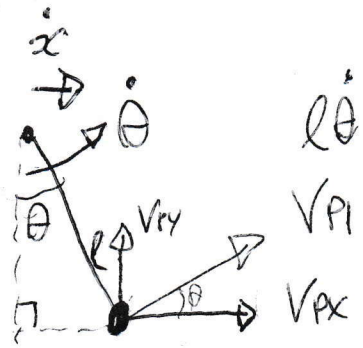
EXAMPLE: MASS - SPRING - PENDULUM.

LAGRANGIAN: $L = K - P$

$$K = K_{CART} + K_{PENDULUM}$$

$$= \frac{1}{2} M_1 \dot{x}^2 + \frac{1}{2} M_2 V_P^2$$

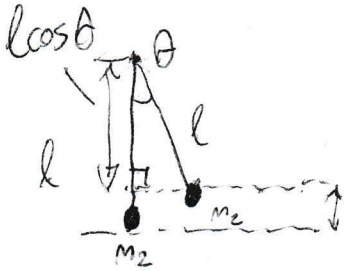


$$V_P^2 = V_{PX}^2 + V_{PY}^2$$

$$= (\dot{x} + \ell\dot{\theta}\cos\theta)^2 + (\ell\dot{\theta}\sin\theta)^2$$

$$\therefore \quad K = \frac{1}{2} M_1 \dot{x}^2 + \frac{1}{2} M_2 \left((\dot{x} + \ell\dot{\theta}\cos\theta)^2 + (\ell\dot{\theta}\sin\theta)^2 \right)$$

$$= \frac{1}{2}(M_1 + M_2)\dot{x}^2 + \frac{1}{2} M_2 \left(\ell^2\dot{\theta}^2 + 2\ell\dot{\theta}\dot{x}\cos\theta \right)$$



$$P = P_{SPRING} + P_{PENDULUM}$$

$$= \frac{1}{2} k x^2 + M_2 g (\ell - \ell\cos\theta)$$

$$= \frac{1}{2} k x^2 + M_2 g \ell (1 - \cos\theta)$$

$$L = K - P = \frac{1}{2}(M_1 + M_2)\dot{x}^2 + \frac{1}{2} M_2 \left(\ell^2\dot{\theta}^2 + 2\ell\dot{\theta}\dot{x}\cos\theta \right) - \frac{1}{2} k x^2 - M_2 g \ell (1 - \cos\theta)$$

ROTATIONAL MOTION

$$T = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta}$$

$$\frac{\partial L}{\partial \dot{\theta}} = M_2 \ell^2 \dot{\theta} + M_2 \ell \dot{x} \cos\theta$$

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = M_2 l^2 \ddot{\theta} + M_2 l \ddot{x} \cos\theta - M_2 l \dot{x} \dot{\theta} \sin\theta$$

$$\frac{\partial L}{\partial \theta} = -M_2 g l \sin\theta - M_2 l \dot{\theta} \dot{x} \sin\theta$$

$$\therefore T = M_2 l^2 \ddot{\theta} + M_2 l \ddot{x} \cos\theta + M_2 g l \sin\theta.$$

TRANSLATIONAL MOTION

$$F = \frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{x}}\right) - \frac{\partial L}{\partial x}$$

$$\frac{\partial L}{\partial \dot{x}} = (M_1 + M_2)\dot{x} + M_2 l \dot{\theta} \cos\theta$$

$$\frac{\partial}{\partial t}\left(\frac{\partial L}{\partial \dot{x}}\right) = (M_1 + M_2)\ddot{x} + M_2 l \ddot{\theta} \cos\theta - M_2 l \dot{\theta}^2 \sin\theta$$

$$\frac{\partial L}{\partial x} = -kx$$

$$\therefore F = (M_1 + M_2)\ddot{x} + M_2 l \ddot{\theta} \cos\theta - M_2 l \dot{\theta}^2 \sin\theta + kx.$$